

SODANKYLA, Finland

WMO: 028360

Lat: 67.367N

Lon: 26.629E

Elev: 184

StdP: 99.14

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.8	-30.8	-36.4	0.1	-33.5	-33.4	0.2	-30.6	8.3	-4.6	7.6	-4.7	1.2	310	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	25.2	16.3	23.0	15.3	20.9	14.5	17.9	22.7	16.6	20.9	15.5	19.4	3.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.7	19.6	14.9	10.8	18.3	13.7	10.0	17.0	51.3	22.5	47.3	20.9	43.9	19.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	6.0	5.3	-37.2	27.7	3.5	2.0	-39.7	29.2	-41.7	30.4	-43.7	31.5	-46.2	33.0		
			-37.0	19.5	3.2	1.5	-39.3	20.5	-41.2	21.4	-43.0	22.3	-45.3	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-13.2	-12.8	-7.7	-1.2	5.3	11.4	14.8	12.3	7.1	0.3	-6.4	-10.3	(6)	
(7)		DBStd	11.29	8.42	8.32	6.25	4.76	4.44	3.95	3.50	3.56	3.73	4.72	6.63	8.17	(7)	
(8)		HDD10.0	3974	718	639	547	336	159	29	3	17	102	302	492	630	(8)	
(9)		HDD18.3	6691	976	872	806	586	405	209	118	191	338	560	742	888	(9)	
(10)		CDD10.0	338	0	0	0	0	12	72	153	87	14	0	0	0	(10)	
(11)		CDD18.3	15	0	0	0	0	0	3	9	3	0	0	0	0	(11)	
(12)		CDH23.3	157	0	0	0	0	6	32	93	26	0	0	0	0	(12)	
(13)		CDH26.7	23	0	0	0	0	0	4	16	3	0	0	0	0	(13)	
(14)	Wind	WSAvg	2.5	2.3	2.4	2.7	2.5	2.7	2.6	2.4	2.4	2.5	2.4	2.3	2.4	(14)	
(15)	Precipitation	PrecAvg	534	35	30	28	29	38	58	72	62	53	49	43	37	(15)	
(16)		PrecMax	784	71	72	66	79	101	113	139	136	116	117	72	77	(16)	
(17)		PrecMin	408	6	4	4	7	4	6	8	19	6	5	8	9	(17)	
(18)		PrecStd	83	15	16	15	15	19	27	31	29	25	23	17	16	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	3.1	6.1	13.5	24.0	27.2	28.6	26.3	18.4	10.8	5.9	3.4	(19)	
(20)		MCWB	1.1	1.6	2.4	7.5	13.6	16.7	17.9	17.9	13.2	9.2	5.3	2.0	2.0	(20)	
(21)		2%	DB	0.4	0.7	3.6	9.6	19.6	23.2	25.9	23.2	16.3	9.3	3.6	1.7	(21)	
(22)		MCWB	-0.1	-0.1	0.9	4.7	11.9	14.4	17.1	15.9	12.1	8.0	3.0	1.1	1.1	(22)	
(23)		5%	DB	-0.9	-0.2	1.7	7.3	15.8	20.9	23.7	20.9	14.6	8.0	2.1	0.6	(23)	
(24)		MCWB	-1.4	-0.8	-0.2	3.4	9.5	13.5	16.0	14.9	11.5	6.8	1.7	0.1	0.1	(24)	
(25)		10%	DB	-2.4	-1.6	0.5	5.5	13.1	18.7	21.7	18.7	13.0	6.3	1.0	-0.4	(25)	
(26)		MCWB	-2.8	-2.2	-0.8	2.2	8.1	12.3	15.3	14.1	10.9	5.4	0.6	-0.8	-0.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.8	2.8	7.7	14.0	18.0	19.7	19.1	14.3	9.9	5.5	2.2	(27)	
(28)		MCDWB	1.9	2.9	5.8	12.8	21.6	23.9	24.1	24.3	16.7	10.4	5.7	2.8	2.8	(28)	
(29)		2%	WB	0.0	0.2	1.2	5.1	12.4	16.0	18.4	17.3	13.0	8.3	3.1	1.2	(29)	
(30)		MCDWB	0.4	0.7	3.0	8.5	18.8	20.9	23.6	21.2	15.2	9.1	3.5	1.6	1.6	(30)	
(31)		5%	WB	-1.3	-0.8	0.3	3.8	10.3	14.3	17.3	15.7	12.1	6.9	1.8	0.2	(31)	
(32)		MCDWB	-1.0	-0.2	1.4	6.7	14.9	19.0	21.7	19.2	14.0	7.9	2.0	0.6	0.6	(32)	
(33)		10%	WB	-2.8	-2.2	-0.7	2.6	8.6	13.1	16.2	14.6	11.0	5.5	0.7	-0.8	(33)	
(34)		MCDWB	-2.4	-1.7	0.5	4.9	12.5	17.5	20.3	17.8	12.7	6.3	0.9	-0.4	-0.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.5	8.9	10.6	9.5	8.8	8.6	8.9	8.9	7.3	5.0	6.0	7.7	7.7	(35)	
(36)		5% DB	MCDBR	6.6	6.2	9.3	10.9	14.3	12.7	12.8	12.6	10.1	5.4	3.8	4.8	4.8	(36)
(37)		MCWBR	6.5	5.7	7.3	6.8	7.5	5.5	5.5	6.2	6.4	4.4	3.8	4.7	4.7	(37)	
(38)		5% WB	MCDBR	6.8	6.1	7.5	9.7	12.7	10.4	10.5	10.3	8.2	4.9	3.7	4.7	4.7	(38)
(39)		MCWBR	6.7	5.7	6.2	6.3	7.2	5.3	5.4	5.7	5.5	4.3	3.7	4.6	4.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.193	0.257	0.291	0.323	0.340	0.341	0.354	0.336	0.310	0.277	0.351	N/A	N/A	(40)	
(41)		taud	1.924	2.093	2.162	2.206	2.422	2.474	2.454	2.483	2.536	2.328	2.710	N/A	N/A	(41)	
(42)		Ebn at Noon	147	556	747	825	857	865	838	817	755	550	180	0	0	(42)	
(43)		Edh at Noon	12	50	82	104	95	93	92	80	60	40	14	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.05	0.55	2.06	3.93	4.82	5.08	4.65	3.40	1.76	0.59	0.11	0.00	0.00	(44)	
(45)		RadStd	0.00	0.08	0.17	0.27	0.43	0.42	0.47	0.32	0.18	0.11	0.01	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.70	N/A	N/A	+0.82	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page